

Structural Inductive Biases in Weight Embeddings for Neural Network Property Prediction

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Abstract

Weight space embeddings enable model selection, editing, and analysis, yet most approaches flatten all parameters into a single vector, discarding architectural structure. We ask: does respecting network structure improve embedding quality for property prediction? Through systematic ablation on the CIFAR-10 Model Zoo ($N = 61,335$ models), we demonstrate that layer-wise encoding—computing per-layer statistics (mean, std, min, max)—significantly outperforms naïve flattening for accuracy prediction. Layer-wise encoding achieves Pearson correlation $r = 0.547$ versus $r = 0.421$ for Flatten+MLP, an improvement of $\Delta r = 0.126$ ($p < 0.001$, $t = 12.85$). The effect holds across all five random seeds with 100% consistency. Our ablation ladder isolates the contribution of structural biases: layer-awareness provides substantial benefit; alignment and equivariance remain to be evaluated at scale. We validate the CIFAR-10 Model Zoo as a viable benchmark ($\sigma = 15.62\%$ accuracy variance) and provide a methodology for fair embedding comparison. Results suggest that structural inductive biases are as important in weight space as in input space—a principle that should guide embedding method design.

1 Introduction

We ask a simple question: does respecting neural network structure improve weight embedding quality? Neural network weights encode rich information about model functionality—training history, learned representations, and performance characteristics. Yet the dominant approach to weight embeddings flattens all parameters into a single vector, discarding the architectural structure that organizes these weights into meaningful layers.

This matters because weight embeddings underpin critical applications: model selection from large zoos, neural network editing, and automated model analysis. When practitioners must choose among tens of thousands of pretrained models, embeddings that accurately predict properties like accuracy enable efficient selection. Poor embeddings waste compute evaluating unsuitable candidates.

The weight embedding literature has developed several approaches with varying structural sophistication. Hyper-Representations use variational autoencoders on flattened weights. Neural Functional Transformers provide permutation-equivariant processing. Git Re-Basin enables weight alignment across permutation symmetries. However, no systematic comparison exists evaluating these methods on the same benchmark with consistent metrics. We cannot answer the basic question: which structural biases actually improve property prediction?

We address this gap through a principled ablation study. We design an **embedding ladder** with increasing structural sophistication:

1. **Flatten+MLP**: Flatten all weights, encode with MLP (no structure)
2. **Layer-wise**: Per-layer statistics with mean aggregation (structure-aware)
3. **Layer-wise+GRB**: Add Git Re-Basin alignment preprocessing
4. **NFN**: Permutation-equivariant architecture

Each step adds exactly one structural inductive bias, isolating its contribution to property prediction.

Our key finding is striking: simply preserving layer boundaries—computing per-layer statistics (mean, std, min, max) rather than flattening—improves accuracy prediction correlation by $\Delta r = 0.126$ (from $r = 0.421$

to $r = 0.547$, $p < 0.001$). This 30% relative improvement holds across all five random seeds tested. The effect size substantially exceeds our conservative threshold of $\Delta r > 0.1$.

Why does layer-wise encoding help? Each neural network layer learns different levels of abstraction. Early layers detect edges; deeper layers encode semantic features. Their weight distributions reflect these functional differences. Flattening destroys this hierarchical structure, mixing edge-detector statistics with semantic-feature statistics. Layer-wise encoding preserves the functional organization.

This paper makes three contributions:

1. **Systematic ablation methodology** for evaluating structural biases in weight embeddings
2. **Quantified evidence** that layer-wise encoding provides substantial improvement over flattening ($\Delta r = 0.126$, $p < 0.001$)
3. **Validated benchmark** confirming CIFAR-10 Model Zoo ($N = 61,335$, $\sigma = 15.62\%$) as viable testbed

We validate on CIFAR-10 Model Zoo, leaving cross-dataset generalization to future work. Our results suggest that structural inductive biases are as important in weight space as they are in input space—a principle that should guide future embedding method design.

2 Related Work

2.1 Weight Space Learning

Neural network weights exist in high-dimensional spaces with rich geometric structure. Linear mode connectivity demonstrates that trained networks can be connected via low-loss paths in weight space, suggesting meaningful structure beyond random initialization. This motivates learning representations that capture weight space semantics.

Hyper-Representations Schürholt et al. [2022b] train variational autoencoders on collections of neural network weights, demonstrating that latent spaces capture training dynamics and model relationships. However, evaluation focused on weight generation rather than property prediction, and comparisons used different datasets and metrics across papers.

Statistical approaches (StatNN and variants) compute simple statistics (mean, variance) over weights for property prediction. While computationally efficient, these methods ignore network structure entirely, treating all parameters as an unordered bag.

2.2 Permutation Symmetries in Weight Space

Neural networks exhibit permutation symmetries: reordering hidden neurons with corresponding weight adjustments preserves function. This symmetry complicates weight space learning—functionally identical networks may have distant representations.

Git Re-Basein ? addresses this through weight alignment, finding permutations that minimize distance between networks. Originally developed for model merging, alignment could preprocess weights before embedding. The computational cost scales quadratically with model count, creating challenges at scale.

Neural Functional Transformers (NFN) Zhou et al. [2024] take an architectural approach, designing layers that are permutation-equivariant by construction. This avoids alignment preprocessing but requires specialized architectures. NFN achieves strong results on weight processing tasks but has not been systematically compared to simpler alternatives for property prediction.

2.3 Model Zoos and Benchmarks

Large collections of pretrained models with metadata enable systematic evaluation. **Model Zoos** Schürholt et al. [2022a] provide thousands of checkpoints with ground-truth accuracy labels, establishing infrastructure for weight embedding benchmarks. However, existing evaluations compare methods on different zoo subsets with incompatible protocols.

2.4 Gap in Existing Work

Prior work evaluates embedding methods in isolation on different datasets with different metrics. We cannot determine which structural biases matter most for property prediction. This paper provides the missing systematic comparison through a controlled ablation study.

3 Methodology

3.1 Ablation Ladder Design

Our goal is isolating the contribution of structural inductive biases to property prediction. We design a four-step ablation ladder where each step adds exactly one bias:

Step	Method	Structural Bias Added
1	Flatten+MLP	None (baseline)
2	Layer-wise	Per-layer processing
3	Layer-wise+GRB	Permutation alignment
4	NFN	Permutation equivariance

Table 1: Ablation ladder with increasing structural sophistication.

3.2 Embedding Architectures

Flatten+MLP (Baseline): All network parameters are concatenated into a single vector and passed through an MLP encoder. This baseline ignores network structure entirely.

Layer-wise Encoding: For each layer l , we compute four statistics: mean, standard deviation, minimum, and maximum of the weight values. These statistics are concatenated across layers and encoded via MLP. This preserves layer-level structure.

Layer-wise+GRB: Before computing layer-wise statistics, we align each model to a reference using Git Re-Basin, removing permutation-induced variance.

Neural Functional Transformer: NFN processes weights using permutation-equivariant layers by design, avoiding explicit alignment.

3.3 Dataset

We use the CIFAR-10 Model Zoo containing $N = 61,335$ pretrained CNN checkpoints with ground-truth accuracy labels. Key properties:

- **Accuracy variance:** $\sigma = 15.62\%$ (above 10% threshold)
- **Split:** 80% train / 20% test, fixed across methods

3.4 Training Protocol

- **Optimizer:** AdamW (lr=0.001, weight_decay=0.0001)
- **Epochs:** 50 with early stopping (patience=10)
- **Seeds:** 5 independent runs

3.5 Evaluation Metrics

- **Primary:** Pearson correlation (r) between predicted and ground-truth accuracy
- **Statistical test:** Paired t -test across seeds ($\alpha = 0.05$)

3.6 Success Criteria

- **P1:** Layer-wise > Flatten+MLP by $\Delta r > 0.1$ ($p < 0.05$)
- **P2:** Layer-wise+GRB > Layer-wise by $\Delta r > 0.05$ ($p < 0.05$)
- **P3:** NFN > Layer-wise+GRB by $\Delta r > 0.05$ ($p < 0.05$)

4 Experimental Setup

4.1 Dataset Details

The CIFAR-10 Model Zoo contains 61,335 CNN checkpoints with accuracy range 7.33%–56.83%, mean 32.12%, and standard deviation 15.62%.

4.2 Methods Evaluated

Method	Status
Flatten+MLP	Completed
Layer-wise	Completed
Layer-wise+GRB	Resource-limited
NFN	Blocked

Table 2: Evaluation status of ablation ladder methods.

4.3 Implementation Details

Layer-wise Encoder: Input $4 \times L$ statistics per layer, hidden 256 units, output 128-dim embedding.

Regressor Head: Input 128-dim, hidden 64 units, output 1 (predicted accuracy).

Training: AdamW optimizer, ReduceLROnPlateau schedule, batch size 256, early stopping.

5 Results

5.1 Dataset Validation (H-E1)

The CIFAR-10 Model Zoo exhibits accuracy $\sigma = 15.62\%$, exceeding our 10% threshold.

Gate H-E1: PASS

5.2 Main Result: Layer-wise vs Flatten (H-M1)

Method	Mean r	Std r
Flatten+MLP	0.421	0.006
Layer-wise	0.547	0.007
Difference	+0.126	—

Table 3: Accuracy prediction correlation across methods.

Statistical Significance: $\Delta r = 0.1262$, $t = 12.847$, $p = 0.0002$

Seed Consistency: 5/5 seeds show improvement (100%)

Gate H-M1: PASS

5.3 Alignment Experiment (H-M2)

Git Re-Basin alignment at 61K scale exceeded computational budget. Implementation verified correct; requires GPU infrastructure.

Gate H-M2: LIMITATION

5.4 Summary

Claim	Status
Layer-wise > Flatten by $\Delta r > 0.1$	SUPPORTED
GRB > Layer-wise	INCONCLUSIVE
NFN > GRB	NOT TESTED

Table 4: Hypothesis evaluation summary.

6 Discussion

6.1 Key Findings Interpretation

Layer-wise encoding improves accuracy prediction by $\Delta r = 0.126$ (30% relative gain) because it preserves functional hierarchy: different layers learn different abstractions, and their weight statistics reflect these differences.

6.2 Incomplete Ablation

GRB alignment at 61K scale was computationally prohibitive. NFN evaluation was blocked by the H-M2 prerequisite. The complete ablation ladder remains partially evaluated.

6.3 Limitations

1. **Single dataset:** CIFAR-10 Model Zoo only
2. **Partial ablation:** 2 of 4 steps completed
3. **CPU-only constraints:** Limited experiment scale

6.4 Broader Implications

Preserve structure when embedding weights. This principle should generalize beyond our specific statistics. Our ablation methodology provides a template for future systematic comparisons.

7 Conclusion

We asked whether respecting neural network structure improves weight embedding quality. The answer is unambiguously yes.

Layer-wise encoding improves accuracy prediction correlation by $\Delta r = 0.126$ ($p < 0.001$) over naïve flattening. This 30% relative improvement holds across all five random seeds.

Three implications:

1. Layer-wise encoding should be the default baseline, not flattening
2. Structural preservation matters in weight space as in input space
3. Controlled ablation enables fair cross-method comparison

The goal is universal weight embeddings generalizing across architectures and properties. Our results suggest structural inductive biases—starting with layer boundaries—are essential ingredients.

References

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